

COURIER MANAGEMENT SYSTEM(CMS)

5. PROJECT DEVELOPMENT PHASE

5.1 Project Development

The **Project Development Phase** is the stage where the planned design and requirements are converted into a working **Courier Management System (CMS)**.

Main objectives:

- Develop the required system modules.
- Create and connect the database.
- Implement courier management features.
- Integrate different modules.
- Test features during development.

5.2 Development Process

The project is developed through the following steps:

Requirement Analysis



System Design



Database Design



Module Development



Integration



Testing & Debugging



Final System

5.3 Database Development

The database is designed to store and manage project information.

Main objects/tables:

- Customer
- Shipment

- Delivery Agent
- Branch
- Delivery Update
- Invoice

Database stores:

- Customer details
- Shipment details
- Agent information
- Delivery status
- Invoice and payment details
- Branch information

5.4 Customer Management Development

The Customer Management module is developed to manage customer information.

Features:

- Customer registration.
- Customer login.
- Add customer details.
- Update customer information.
- Search customer records.
- View shipment history.

5.5 Shipment Management Development

The Shipment module manages the complete shipment creation process.

Features:

- Enter sender details.
- Enter receiver details.
- Enter parcel details.
- Enter pickup and delivery addresses.

- Generate tracking ID.
- Store shipment information.

Flow:

Customer Input → Shipment Creation → Tracking ID

5.6 Delivery Agent Development

This module manages delivery agents and their assignments.

Features:

- Add delivery agent details.
- View available agents.
- Assign shipments.
- Manage delivery schedules.
- Update agent availability.

Flow:

Shipment → Available Agent → Agent Assignment

5.7 Shipment Tracking Development

The tracking module allows users to monitor shipment progress.

Shipment statuses:

- Booked
- Picked Up
- In Transit
- Out for Delivery
- Delivered

The delivery agent updates the status, and the latest information is made available to customers and staff.

5.8 Invoice Development

The Invoice module manages courier charges and payment information.

Features:

- Calculate delivery charges.
- Generate invoice.
- Store invoice details.

- Record payment status.
- View invoice information.

Flow:

Shipment Details → Charge Calculation → Invoice → Payment Status

5.9 Reports and Dashboard Development

The reporting module provides useful information for administrators and managers.

Reports:

- Shipment Report
- Delivery Report
- Revenue Report
- Agent Performance Report
- Branch-wise Report

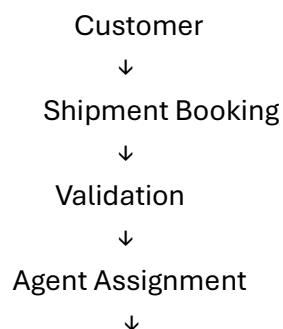
Dashboard:

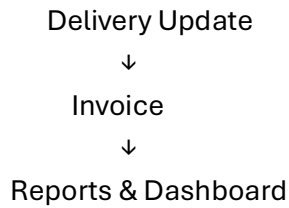
- Total Shipments
- Delivered Shipments
- Pending Shipments
- Revenue Overview
- Agent Performance
- Branch Performance

5.10 Module Integration

All modules are connected to work as a single system.

Integration Flow:





- This integration ensures that information entered in one module can be used by other related modules.

5.11 Development Outcome

After completing the development phase:

- Customer management is implemented.
- Shipment booking is available.
- Delivery agents can be assigned.
- Shipment status can be updated.
- Invoices can be generated.
- Reports and dashboards are available.
- Modules are integrated into one system.